PRATIK NAIR

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PROFESSIONAL SUMMARY

Python developer & AI test engineer with experience at HARMAN International and Iraje Sofwtares in LLM prompt engineering, automotive cybersecurity validation, and test automation. Specializes in building AI-driven QA frameworks for embedded automotive systems; experienced across CI/CD, AOSP, and ISO/SAE 21434-aligned cybersecurity domain

WORK EXPERIENCE

Associate Software Engineer March 2025 - Present
Harman International, Bengaluru

- Co-developed an **MCP-based Android HMI automation agent** (Python) as part of a team, contributing to the design and implementation of **30+ MCP tools** covering screen parsing **(parallel OCR + icon detection)**, Set-of-Mark visual grounding, **HSV-based** UI state inspection, **BFS state-graph** navigation, and semantic test recording with auto-generated Python replay scripts.
- Engineered **50+ LLM prompts** for generative AI-assisted test script generation, reducing test documentation time by **∼60%** — cutting per-API scripting from **1 hour to under 15 minutes** for multiple interfaces simultaneously.
- Pioneered interface test automation across **7 domains**, authoring **720+ JUnit 5 (Java) and GTest (C++)** test cases by interpreting **40+ Architecture Planning Items (APIs) and sequence diagrams** — first-ever interface testing initiative on the project.
- Built and deployed a **Jenkins CI/CD pipeline on AWS** (Gradle, lcov); cut build time by eliminating redundant repository pulls and optimizing source fetch strategies.
- Validated cybersecurity features across **Gateway ECU, TCU, and HSM** — covering encrypted diagnostic logging, VLAN-based network isolation, secure OTA pipelines, and negative-path build image tampering; verified **50+ builds** aligned with **ISO/SAE 21434** and **UNECE WP.29/R155**.
- Developed **Python automation scripts** (PyAutoGUI, NumPy, soundcard) for Android AOSP emulators, saving **20 hours of manual testing per sprint**; executed CTS across **847 modules (2.2M+ test cases, 99.97% pass rate)** and VTS across **399 modules (18,668 test cases)**.
- Triaged **3–4 defects daily** with root cause analysis; drove **2 regression cycles per release build** on embedded systems spanning CAN, UDS, Bluetooth, and firmware update validation.

Cybersecurity Developer August 2024 - March 2025
Iraje Software, Mumbai

- Built and client-deployed a Linux-based **real-time login intrusion detection system** in Python (**∼680 LOC**); monitored auth logs **24/7** via a systemd service, alerting admins over **TCP** to a centralized **Oracle 11G** database with zero downtime post-deployment.

PROJECTS

LockBox *FastAPI, Python, JWT*
Async authentication microservice (FastAPI, Python) with 6 REST endpoints covering full auth lifecycle; implemented JWT access/refresh token pair with DB-backed revocation on logout, bcrypt password hashing, and rate limiting (5 req/min/IP) to mitigate brute-force attacks

Imarec *Python, TensorFlow, Keras, Flask*
CNN image classifier on **CIFAR-10** (60K images, 10 classes); achieved **∼82% test accuracy** with **∼80ms inference time** per image via a Flask web interface.

PixelDetect *Python, Flask, OpenCV, MobileNet SSD*
Real-time object detection web app achieving **∼20 FPS** on live video; detects **80 COCO object classes** at **0.5 confidence threshold** with bounding-box overlay.

EDUCATION

Master's in Computer Applications Hiray Institute Of Computer Applications, Mumbai 2025
8.5 CGPA

Bachelor's in Information Technology, Thakur College Of Science and Commerce, Mumbai 2023
9.1 CGPA

HSC, Bhavan's College, Mumbai 2020
62.15% (Science)

SSC, St John The Evangelist High School, Mumbai 2018
86.8%

CERTIFICATIONS

Understanding LLMs and Basic Prompting Techniques – CodeSignal (2025)
AI Governance Certification – Securiti AI (2025)
Introduction to Jenkins – Linux Foundation (2025)
Cloud Computing – NPTEL (2024) | **Top 5% among 24,000+ candidates** | Score: 83%

SKILLS

Languages & Frameworks: Python, Java, C++, Flask, TensorFlow, Keras, OpenCV
AI / ML / LLMs: Prompt Engineering, LLM Integration, CNN, MobileNet SSD, OCR, Computer Vision
Test & Automation: JUnit 5, GTest, PyAutoGUI, ADB, AOSP, CTS/VTS, API Mocking
DevOps & Cloud: Jenkins, AWS, Gradle, Git, GitHub, lcov, CI/CD Pipelines
Automotive & Security: CAN, UDS, ISO/SAE 21434, UNECE WP.29, Secure Boot, IDS, OTA Validation
Databases: Oracle 11G, MySQL, Firebase
OS & Tools: Linux, Bash, Jira, ALM, Confluence